

✓ 4. Nagle's algorithm causes increased latency primarily because it:

*1/1

- ☐ a) Disables ACK piggybacking
- ☒ b) Buffers small segments until ACK arrives ✓
- ☐ c) Reduces CWND on every packet
- ☐ d) Increases SYN backlog

✓ 6. UDP is preferred over TCP for DNS queries primarily because:

*1/1

- ☐ a) UDP supports fragmentation
- ☒ b) DNS responses are small and connection overhead is unnecessary ✓
- ☐ c) TCP cannot use port 53
- ☐ d) UDP has better error correction

✓ 7.A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is: *1/1

- ☐ a) Increase UDP buffer size
- ☐ b) Switch to TCP
- ☒ c) Implement Forward Error Correction (FEC) ✓
- ☐ d) Reduce frame rate to 1 fps

✓ 14.In HTTP caching, the Cache-Control: no-cache directive means: *1/1

- ☐ a) Never cache this resource
- ☒ b) Validate with the server before using a cached copy ✓
- ☐ c) Cache for 0 seconds
- ☐ d) Disable all caching headers

Name of the student-Reg.No *

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✓ 10.HTTP/1.1 persistent connections reduce latency by: *1/1

- ☐ a) Enabling UDP fallback
- ☒ b) Reusing the same TCP connection for multiple requests ✓
- ☐ c) Compressing HTTP headers
- ☐ d) Sending requests without waiting for DNS

✓ 11.Head-of-line blocking in HTTP/1.1 occurs when: *1/1

- ☐ a) The DNS server is slow
- ☒ b) A response blocks subsequent responses on the same connection ✓
- ☐ c) TCP CWND drops to zero
- ☐ d) The server rejects pipelined requests



✓ 18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases: *1/1

- ☐ a) Cache hit ratio
- ☒ b) Recursive resolver query load ✓
- ☐ c) DNSSEC signature validity
- ☐ d) Zone transfer frequency

✓ 19. A user browses to a new website. The correct sequence of protocol interactions for the first byte of the page is: *1/1

- ☐ a) HTTP, TCP, DNS
- ☒ b) DNS, TCP handshake, HTTP GET ✓
- ☐ c) TCP, DNS, HTTP
- ☐ d) DNS, HTTP, TCP

✓ 3. Which field in the TCP header is used to implement flow control at the receiver side? *1/1

- ☐ a) Sequence number
- ☐ b) Acknowledgement number
- ☒ c) Window size ✓
- ☐ d) Urgent pointer

✓ 20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game? *1/1

- ☐ a) TCP for game state, HTTP for assets
- ☒ b) UDP with application-layer sequencing for game state, HTTP/3 for assets ✓
- ☐ c) HTTP/2 for all traffic
- ☐ d) TCP with Nagle enabled for all traffic

CSA0720-MCQ

Total points **19/20** ?

CO3-AT3

✓ 16.A DNS resolver switches from UDP to TCP when: *1/1

- ☐ a) The query type is MX
- ☒ b) The response is truncated (TC bit set) ✓
- ☐ c) TTL exceeds 3600 s
- ☐ d) The authoritative server is unreachable

✓ 15.Which DNS record type maps a domain name to an IPv6 address? *1/1

- ☐ a) A
- ☐ b) CNAME
- ☒ c) AAAA ✓
- ☐ d) MX



✗ 2. A TCP sender has $CWND = 8$ MSS and $SSTHRESH = 16$. After one RTT with no loss, $CWND$ becomes: *0/1

☐ a) 9 MSS

☒ b) 16 MSS ✗

☐ c) 12 MSS

☐ d) 8 MSS

Correct answer

☒ a) 9 MSS

✓ 5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by: *1/1

☐ a) MTU size

☒ b) $CWND$ growth rate relative to RTT ✓

☐ c) DNS resolution time

☐ d) UDP competition

